

Proyecto 2_Entrega 1

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```
datos <- read.csv("test.csv")
submi <- read.csv("sample_submission.csv")

datos <- merge(datos, submi, by="Id")

datos_train <- read.csv("train.csv")

datos_train_uni <- datos_train[, c("LotArea", "SalePrice")]

head(datos)
```

##	Id	MSSubClass	MSZoning	LotFrontage	LotArea	Street	Alley	LotShape
## 1	1461	20	RH	80	11622	Pave	<NA>	Reg
## 2	1462	20	RL	81	14267	Pave	<NA>	IR1
## 3	1463	60	RL	74	13830	Pave	<NA>	IR1
## 4	1464	60	RL	78	9978	Pave	<NA>	IR1
## 5	1465	120	RL	43	5005	Pave	<NA>	IR1
## 6	1466	60	RL	75	10000	Pave	<NA>	IR1
##	LandContour	Utilities	LotConfig	LandSlope	Neighborhood	Condition1	Condition2	
## 1	Lv1	AllPub	Inside	Gtl	NAMES	Feedr	Norm	
## 2	Lv1	AllPub	Corner	Gtl	NAMES	Norm	Norm	
## 3	Lv1	AllPub	Inside	Gtl	Gilbert	Norm	Norm	
## 4	Lv1	AllPub	Inside	Gtl	Gilbert	Norm	Norm	
## 5	HLS	AllPub	Inside	Gtl	StoneBr	Norm	Norm	
## 6	Lv1	AllPub	Corner	Gtl	Gilbert	Norm	Norm	
##	BldgType	HouseStyle	OverallQual	OverallCond	YearBuilt	YearRemodAdd	RoofStyle	
## 1	1Fam	1Story	5	6	1961	1961	Gable	
## 2	1Fam	1Story	6	6	1958	1958	Hip	
## 3	1Fam	2Story	5	5	1997	1998	Gable	
## 4	1Fam	2Story	6	6	1998	1998	Gable	
## 5	TwnhsE	1Story	8	5	1992	1992	Gable	
## 6	1Fam	2Story	6	5	1993	1994	Gable	
##	RoofMatl	Exterior1st	Exterior2nd	MasVnrType	MasVnrArea	ExterQual	ExterCond	
## 1	CompShg	VinylSd	VinylSd	None	0	TA	TA	
## 2	CompShg	Wd Sdng	Wd Sdng	BrkFace	108	TA	TA	
## 3	CompShg	VinylSd	VinylSd	None	0	TA	TA	
## 4	CompShg	VinylSd	VinylSd	BrkFace	20	TA	TA	
## 5	CompShg	HdBoard	HdBoard	None	0	Gd	TA	
## 6	CompShg	HdBoard	HdBoard	None	0	TA	TA	
##	Foundation	BsmtQual	BsmtCond	BsmtExposure	BsmtFinType1	BsmtFinSF1		
## 1	CBlock	TA	TA	No	Rec	468		
## 2	CBlock	TA	TA	No	ALQ	923		
## 3	PConc	Gd	TA	No	GLQ	791		
## 4	PConc	TA	TA	No	GLQ	602		
## 5	PConc	Gd	TA	No	ALQ	263		
## 6	PConc	Gd	TA	No	Unf	0		
##	BsmtFinType2	BsmtFinSF2	BsmtUnfSF	TotalBsmtSF	Heating	HeatingQC	CentralAir	
## 1	LwQ	144	270	882	GasA	TA	Y	
## 2	Unf	0	406	1329	GasA	TA	Y	
## 3	Unf	0	137	928	GasA	Gd	Y	
## 4	Unf	0	324	926	GasA	Ex	Y	
## 5	Unf	0	1017	1280	GasA	Ex	Y	
## 6	Unf	0	763	763	GasA	Gd	Y	
##	Electrical	X1stFlrSF	X2ndFlrSF	LowQualFinSF	GrLivArea	BsmtFullBath		
## 1	SBrkr	896	0	0	896	0		
## 2	SBrkr	1329	0	0	1329	0		
## 3	SBrkr	928	701	0	1629	0		
## 4	SBrkr	926	678	0	1604	0		
## 5	SBrkr	1280	0	0	1280	0		
## 6	SBrkr	763	892	0	1655	0		
##	BsmtHalfBath	FullBath	HalfBath	BedroomAbvGr	KitchenAbvGr	KitchenQual		
## 1	0	1	0	2	1	TA		
## 2	0	1	1	3	1	Gd		

```

## 3      0      2      1      3      1      TA
## 4      0      2      1      3      1      Gd
## 5      0      2      0      2      1      Gd
## 6      0      2      1      3      1      TA
##  TotRmsAbvGrd Functional Fireplaces FireplaceQu GarageType GarageYrBlt
## 1          5      Typ          0      <NA>    Attchd      1961
## 2          6      Typ          0      <NA>    Attchd      1958
## 3          6      Typ          1      TA      Attchd      1997
## 4          7      Typ          1      Gd      Attchd      1998
## 5          5      Typ          0      <NA>    Attchd      1992
## 6          7      Typ          1      TA      Attchd      1993
##  GarageFinish GarageCars GarageArea GarageQual GarageCond PavedDrive
## 1      Unf          1      730      TA      TA      Y
## 2      Unf          1      312      TA      TA      Y
## 3      Fin          2      482      TA      TA      Y
## 4      Fin          2      470      TA      TA      Y
## 5      RFn          2      506      TA      TA      Y
## 6      Fin          2      440      TA      TA      Y
##  WoodDeckSF OpenPorchSF EnclosedPorch X3SsnPorch ScreenPorch PoolArea PoolQC
## 1      140          0          0          0      120      0 <NA>
## 2      393         36          0          0          0      0 <NA>
## 3      212         34          0          0          0      0 <NA>
## 4      360         36          0          0          0      0 <NA>
## 5          0        82          0          0      144      0 <NA>
## 6      157         84          0          0          0      0 <NA>
##  Fence MiscFeature MiscVal MoSold YrSold SaleType SaleCondition SalePrice
## 1 MnPrv      <NA>          0      6  2010      WD      Normal  169277.1
## 2 <NA>      Gar2    12500      6  2010      WD      Normal  187758.4
## 3 MnPrv      <NA>          0      3  2010      WD      Normal  183583.7
## 4 <NA>      <NA>          0      6  2010      WD      Normal  179317.5
## 5 <NA>      <NA>          0      1  2010      WD      Normal  150730.1
## 6 <NA>      <NA>          0      4  2010      WD      Normal  177151.0

```

```
head(datos_train)
```

##	Id	MSSubClass	MSZoning	LotFrontage	LotArea	Street	Alley	LotShape	LandContour
## 1	1	60	RL	65	8450	Pave	<NA>	Reg	Lvl
## 2	2	20	RL	80	9600	Pave	<NA>	Reg	Lvl
## 3	3	60	RL	68	11250	Pave	<NA>	IR1	Lvl
## 4	4	70	RL	60	9550	Pave	<NA>	IR1	Lvl
## 5	5	60	RL	84	14260	Pave	<NA>	IR1	Lvl
## 6	6	50	RL	85	14115	Pave	<NA>	IR1	Lvl
##	Utilities	LotConfig	LandSlope	Neighborhood	Condition1	Condition2	BldgType		
## 1	AllPub	Inside	Gtl	CollgCr	Norm	Norm	1Fam		
## 2	AllPub	FR2	Gtl	Veenker	Feedr	Norm	1Fam		
## 3	AllPub	Inside	Gtl	CollgCr	Norm	Norm	1Fam		
## 4	AllPub	Corner	Gtl	Crawfor	Norm	Norm	1Fam		
## 5	AllPub	FR2	Gtl	NoRidge	Norm	Norm	1Fam		
## 6	AllPub	Inside	Gtl	Mitchel	Norm	Norm	1Fam		
##	HouseStyle	OverallQual	OverallCond	YearBuilt	YearRemodAdd	RoofStyle	RoofMatl		
## 1	2Story	7	5	2003	2003	Gable	CompShg		
## 2	1Story	6	8	1976	1976	Gable	CompShg		
## 3	2Story	7	5	2001	2002	Gable	CompShg		
## 4	2Story	7	5	1915	1970	Gable	CompShg		
## 5	2Story	8	5	2000	2000	Gable	CompShg		
## 6	1.5Fin	5	5	1993	1995	Gable	CompShg		
##	Exterior1st	Exterior2nd	MasVnrType	MasVnrArea	ExterQual	ExterCond	Foundation		
## 1	VinylSd	VinylSd	BrkFace	196	Gd	TA	PConc		
## 2	MetalSd	MetalSd	None	0	TA	TA	CBlock		
## 3	VinylSd	VinylSd	BrkFace	162	Gd	TA	PConc		
## 4	Wd Sdng	Wd Shng	None	0	TA	TA	BrkTil		
## 5	VinylSd	VinylSd	BrkFace	350	Gd	TA	PConc		
## 6	VinylSd	VinylSd	None	0	TA	TA	Wood		
##	BsmtQual	BsmtCond	BsmtExposure	BsmtFinType1	BsmtFinSF1	BsmtFinType2			
## 1	Gd	TA	No	GLQ	706	Unf			
## 2	Gd	TA	Gd	ALQ	978	Unf			
## 3	Gd	TA	Mn	GLQ	486	Unf			
## 4	TA	Gd	No	ALQ	216	Unf			
## 5	Gd	TA	Av	GLQ	655	Unf			
## 6	Gd	TA	No	GLQ	732	Unf			
##	BsmtFinSF2	BsmtUnfSF	TotalBsmtSF	Heating	HeatingQC	CentralAir	Electrical		
## 1	0	150	856	GasA	Ex	Y	SBrkr		
## 2	0	284	1262	GasA	Ex	Y	SBrkr		
## 3	0	434	920	GasA	Ex	Y	SBrkr		
## 4	0	540	756	GasA	Gd	Y	SBrkr		
## 5	0	490	1145	GasA	Ex	Y	SBrkr		
## 6	0	64	796	GasA	Ex	Y	SBrkr		
##	X1stFlrSF	X2ndFlrSF	LowQualFinSF	GrLivArea	BsmtFullBath	BsmtHalfBath	FullBath		
## 1	856	854	0	1710	1	0	2		
## 2	1262	0	0	1262	0	1	2		
## 3	920	866	0	1786	1	0	2		
## 4	961	756	0	1717	1	0	1		
## 5	1145	1053	0	2198	1	0	2		
## 6	796	566	0	1362	1	0	1		
##	HalfBath	BedroomAbvGr	KitchenAbvGr	KitchenQual	TotRmsAbvGrd	Functional			
## 1	1	3	1	Gd	8	Typ			
## 2	0	3	1	TA	6	Typ			

```

## 3      1      3      1      Gd      6      Typ
## 4      0      3      1      Gd      7      Typ
## 5      1      4      1      Gd      9      Typ
## 6      1      1      1      TA      5      Typ
##  Fireplaces FireplaceQu GarageType GarageYrBlt GarageFinish GarageCars
## 1      0      <NA>   Attchd      2003      RFn      2
## 2      1      TA     Attchd      1976      RFn      2
## 3      1      TA     Attchd      2001      RFn      2
## 4      1      Gd     Detchd      1998      Unf      3
## 5      1      TA     Attchd      2000      RFn      3
## 6      0      <NA>   Attchd      1993      Unf      2
##  GarageArea GarageQual GarageCond PavedDrive WoodDeckSF OpenPorchSF
## 1      548      TA      TA      Y      0      61
## 2      460      TA      TA      Y      298     0
## 3      608      TA      TA      Y      0      42
## 4      642      TA      TA      Y      0      35
## 5      836      TA      TA      Y      192     84
## 6      480      TA      TA      Y      40      30
##  EnclosedPorch X3SsnPorch ScreenPorch PoolArea PoolQC Fence MiscFeature
## 1      0      0      0      0      <NA> <NA> <NA>
## 2      0      0      0      0      <NA> <NA> <NA>
## 3      0      0      0      0      <NA> <NA> <NA>
## 4      272     0      0      0      <NA> <NA> <NA>
## 5      0      0      0      0      <NA> <NA> <NA>
## 6      0      320    0      0      <NA> MnPrv  Shed
##  MiscVal MoSold YrSold SaleType SaleCondition SalePrice
## 1      0      2      2008      WD      Normal  208500
## 2      0      5      2007      WD      Normal  181500
## 3      0      9      2008      WD      Normal  223500
## 4      0      2      2006      WD      Abnorml  140000
## 5      0      12     2008      WD      Normal  250000
## 6      700     10     2009      WD      Normal  143000

```

Se selecciona la variable LotArea como la variable a medir. Se puede inferir que entre más area hay, más espacio ya sea para expandir o para realizar actividades puedan atraer a más personas por lo cual puede aumentar su precio de venta.

```

fitLMPW<-lm(LotArea~SalePrice, data = datos_train_uni)
summary(fitLMPW)

```

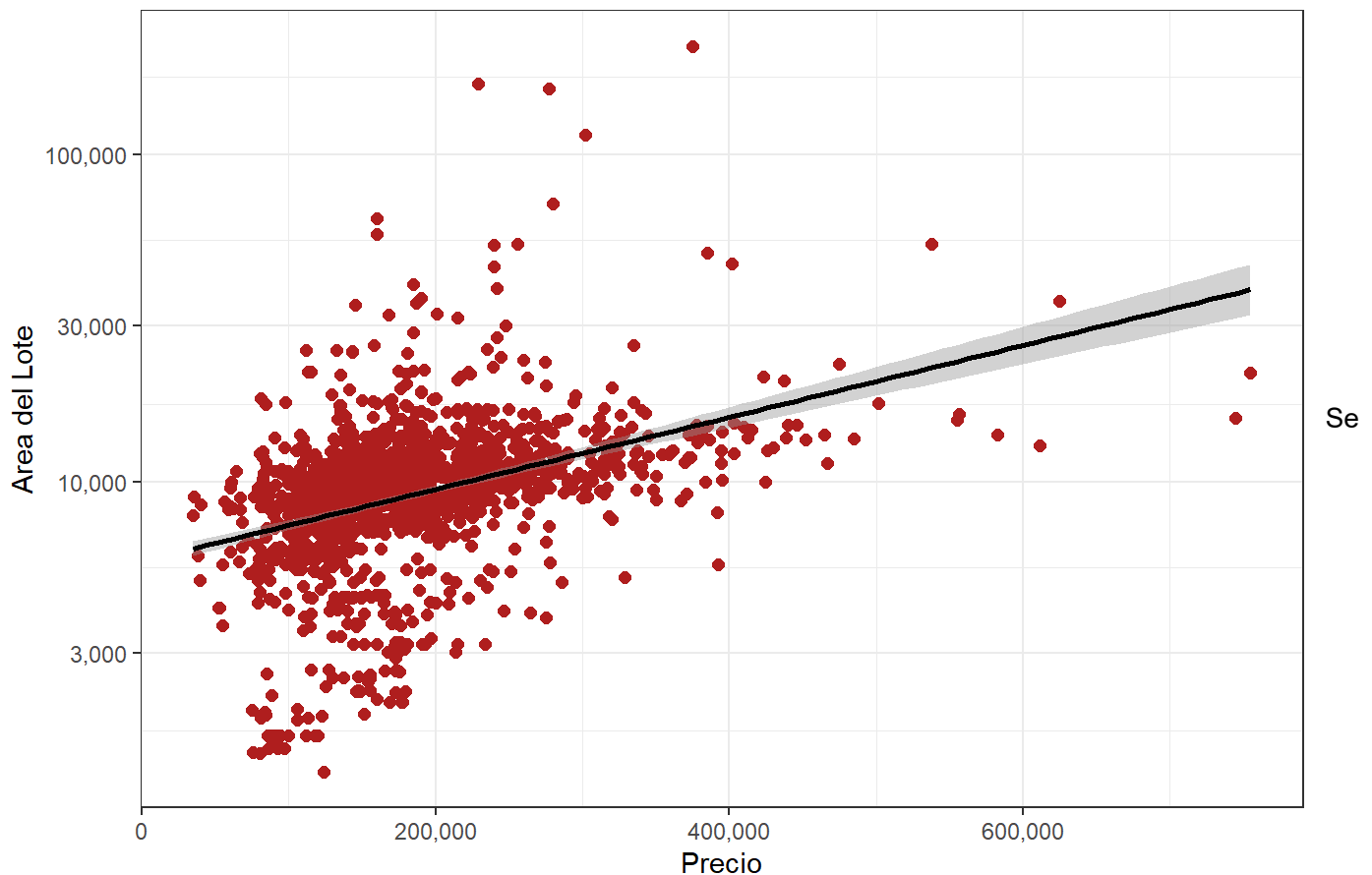
```
##
## Call:
## lm(formula = LotArea ~ SalePrice, data = datos_train_uni)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -13593  -2751   -914   1140  198295
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 4.519e+03  6.271e+02   7.207 9.16e-13 ***
## SalePrice   3.315e-02  3.174e-03  10.445 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9631 on 1458 degrees of freedom
## Multiple R-squared:  0.06961,    Adjusted R-squared:  0.06898
## F-statistic: 109.1 on 1 and 1458 DF,  p-value: < 2.2e-16
```

Se puede inferir en este resumen donde el R^2 ajustado es menor a 1% lo cual puede tener una mayor relación aunque observamos que el otro R^2 es de 0.06 donde puede explicar una porción de la variabilidad del precio de venta.

```
ggplot(data = datos_train_uni, mapping = aes(x = SalePrice, y = LotArea)) +
  geom_point(color = "firebrick", size = 2) +
  geom_smooth(method = "lm", se = TRUE, color = "black") +
  labs(title = "Precio En Venta ~ Estilo De La Casa", x = "Precio", y = "Area del Lote")+
  scale_x_continuous(labels = scales::comma) + # Format large numbers with commas
  scale_y_log10(labels = scales::comma) +      # Use log scale for large LotArea value
  theme_bw() + theme(plot.title = element_text(hjust = 0.5))
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

Precio En Venta ~ Estilo De La Casa



puede observar que la linea de tendencia se dirige hacia arriba. Esto puede significar que entre más area, más aumenta el precio. por lo tanto asumimos que toda la concentración de puntos son casas con menor area y con un precio menor. Por otro lado tambien observamos variabilidad con los datos donde se acercan a la linea lo cual indica que otras variables pueden afectar con el precio de venta de la casa.

Declarando las variable numericas del dataset

```
datos <- datos[, c("MSSubClass", "LotArea", "OverallQual", "OverallCond",
  "YearBuilt", "YearRemodAdd", "BsmtFinSF1", "BsmtFinSF2",
  "BsmtUnfSF", "X1stFlrSF", "X2ndFlrSF", "LowQualFinSF",
  "BsmtFullBath", "BsmtHalfBath", "FullBath", "HalfBath",
  "BedroomAbvGr", "KitchenAbvGr", "TotRmsAbvGrd", "Fireplaces",
  "GarageCars", "GarageArea", "WoodDeckSF", "OpenPorchSF", "EnclosedPorch",
  "X3SsnPorch", "ScreenPorch", "PoolArea", "MiscVal", "MoSold", "YrSold",
  "SalePrice")]

datos_train <- datos_train[, c("MSSubClass", "LotArea", "OverallQual", "OverallCond",
  "YearBuilt", "YearRemodAdd", "BsmtFinSF1", "BsmtFinSF2",
  "BsmtUnfSF", "X1stFlrSF", "X2ndFlrSF", "LowQualFinSF",
  "BsmtFullBath", "BsmtHalfBath", "FullBath", "HalfBath",
  "BedroomAbvGr", "KitchenAbvGr", "TotRmsAbvGrd", "Fireplaces",
  "GarageCars", "GarageArea", "WoodDeckSF", "OpenPorchSF", "EnclosedPorch",
  "X3SsnPorch", "ScreenPorch", "PoolArea", "MiscVal", "MoSold", "YrSold",
  "SalePrice")]
```

```
fitMLM_varSales<-lm(SalePrice~.,data = datos)
```

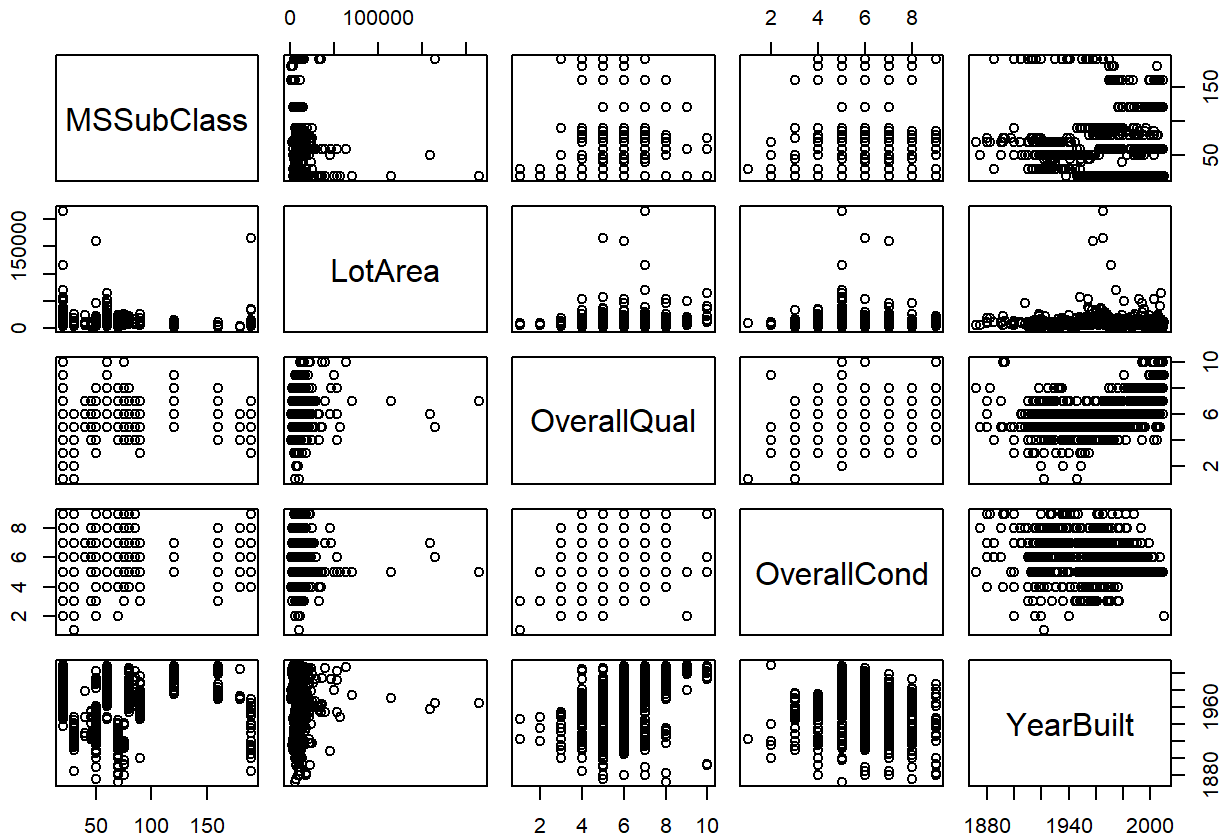
```
summary(fitMLM_varSales)
```



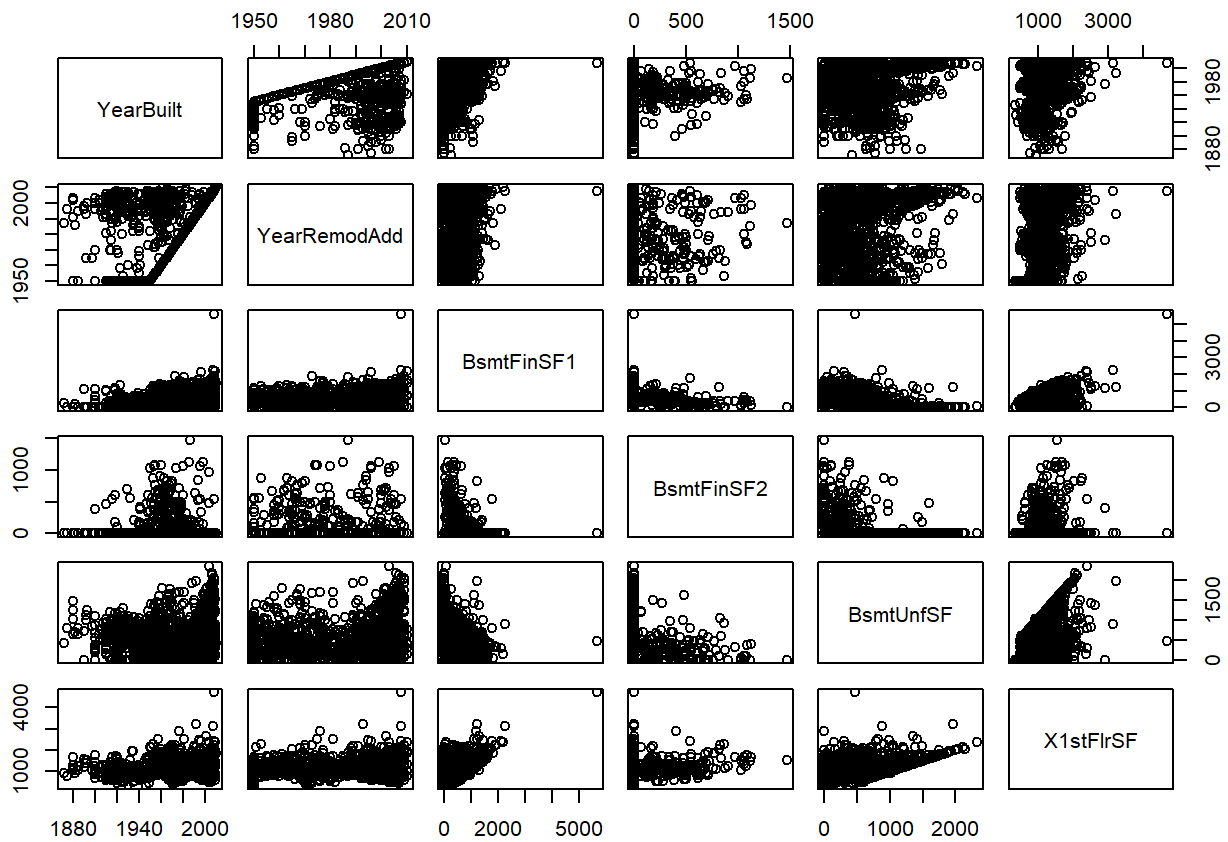
```
##
## Call:
## lm(formula = SalePrice ~ ., data = datos)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.040e-09 -3.340e-10 -4.300e-11  2.460e-10  7.049e-08
##
## Coefficients:
##              Estimate Std. Error  t value Pr(>|t|)
## (Intercept)   1.918e+06  7.980e-08  2.403e+13 < 2e-16 ***
## MSSubClass    -4.103e-13  1.448e-12 -2.830e-01  0.777
## LotArea       1.968e+00  1.319e-14  1.492e+14 < 2e-16 ***
## OverallQual   3.375e-11  6.296e-11  5.360e-01  0.592
## OverallCond   2.462e-11  5.553e-11  4.430e-01  0.658
## YearBuilt     2.678e-12  3.190e-12  8.390e-01  0.401
## YearRemodAdd  -4.410e-12  3.685e-12 -1.197e+00  0.232
## BsmtFinSF1    1.957e-13  2.548e-13  7.680e-01  0.442
## BsmtFinSF2    2.623e-13  3.699e-13  7.090e-01  0.478
## BsmtUnfSF     5.349e-14  2.246e-13  2.380e-01  0.812
## X1stFlrSF    -5.068e-13  3.138e-13 -1.615e+00  0.107
## X2ndFlrSF    -1.479e-13  2.872e-13 -5.150e-01  0.607
## LowQualFinSF  -1.524e-13  1.181e-12 -1.290e-01  0.897
## BsmtFullBath  -1.744e-10  1.383e-10 -1.261e+00  0.207
## BsmtHalfBath  -2.618e-10  2.195e-10 -1.192e+00  0.233
## FullBath      5.045e-11  1.502e-10  3.360e-01  0.737
## HalfBath     -4.445e-11  1.485e-10 -2.990e-01  0.765
## BedroomAbvGr  1.328e+04  9.299e-11  1.428e+14 < 2e-16 ***
## KitchenAbvGr  1.220e-10  3.081e-10  3.960e-01  0.692
## TotRmsAbvGrd  4.199e-11  6.723e-11  6.250e-01  0.532
## Fireplaces   -4.133e-11  9.744e-11 -4.240e-01  0.672
## GarageCars   -6.552e-10  1.617e-10 -4.051e+00  5.39e-05 ***
## GarageArea    2.466e-12  5.610e-13  4.396e+00  1.19e-05 ***
## WoodDeckSF    3.203e-13  4.445e-13  7.210e-01  0.471
## OpenPorchSF   -4.895e-13  8.243e-13 -5.940e-01  0.553
## EnclosedPorch -2.966e-13  8.387e-13 -3.540e-01  0.724
## X3SsnPorch    -4.074e-13  2.523e-12 -1.610e-01  0.872
## ScreenPorch   1.406e-12  9.258e-13  1.518e+00  0.129
## PoolArea      6.605e-13  1.761e-12  3.750e-01  0.708
## MiscVal      -8.216e-15  8.533e-14 -9.600e-02  0.923
## MoSold        1.105e+03  1.899e-11  5.818e+13 < 2e-16 ***
## YrSold        -8.977e+02  3.966e-11 -2.264e+13 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.916e-09 on 1424 degrees of freedom
## (3 observations deleted due to missingness)
## Multiple R-squared:  1, Adjusted R-squared:  1
## F-statistic: 3.398e+27 on 31 and 1424 DF, p-value: < 2.2e-16
```

Se observa principalmente el R^2 ajustado y el multiple tienen un valor de 1 lo cual puede indicar un ajuste perfecto con los variables o puede ser que se analizaron muchas variables lo cual pueden ser irrelevantes como se explica en las siguientes gráficas. Además, al tener un ajuste perfecto puede indicar overfitting de las variables.

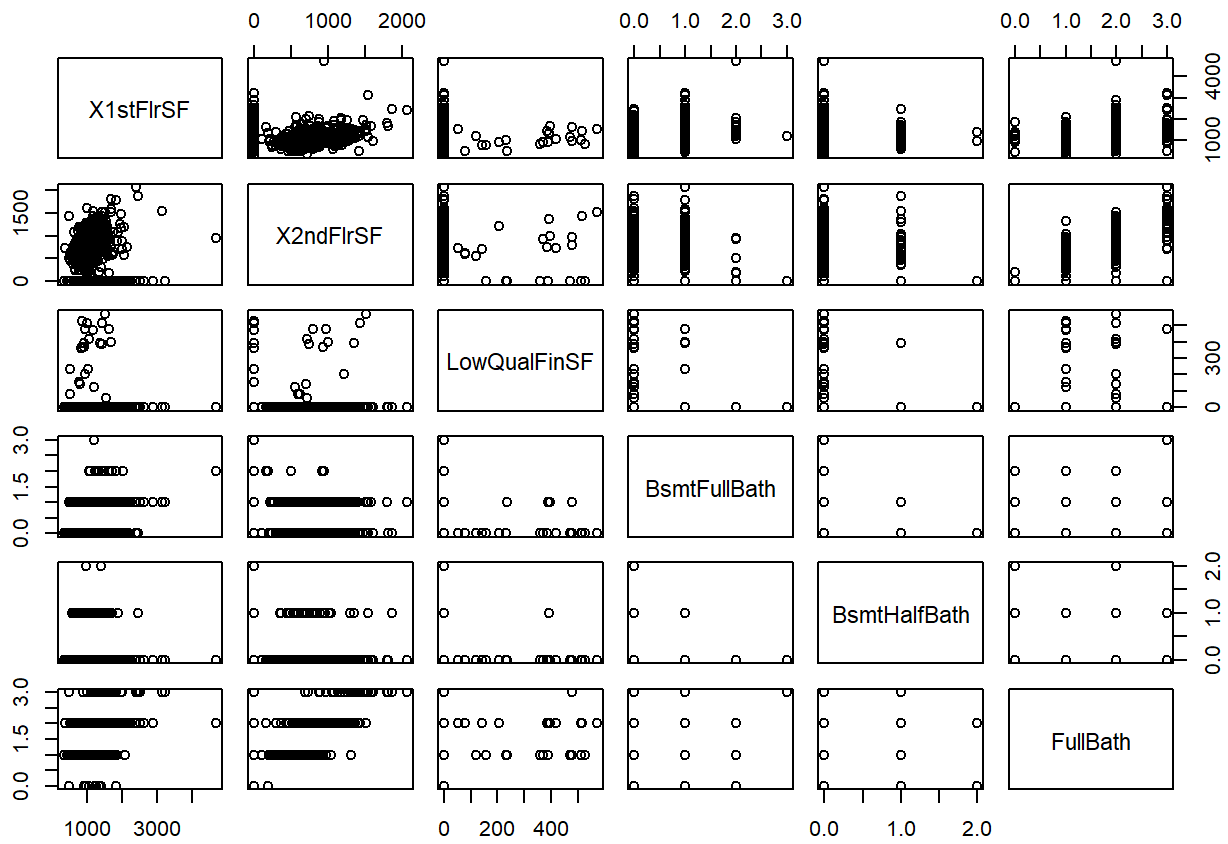
```
plot(datos_train[,1:5])
```



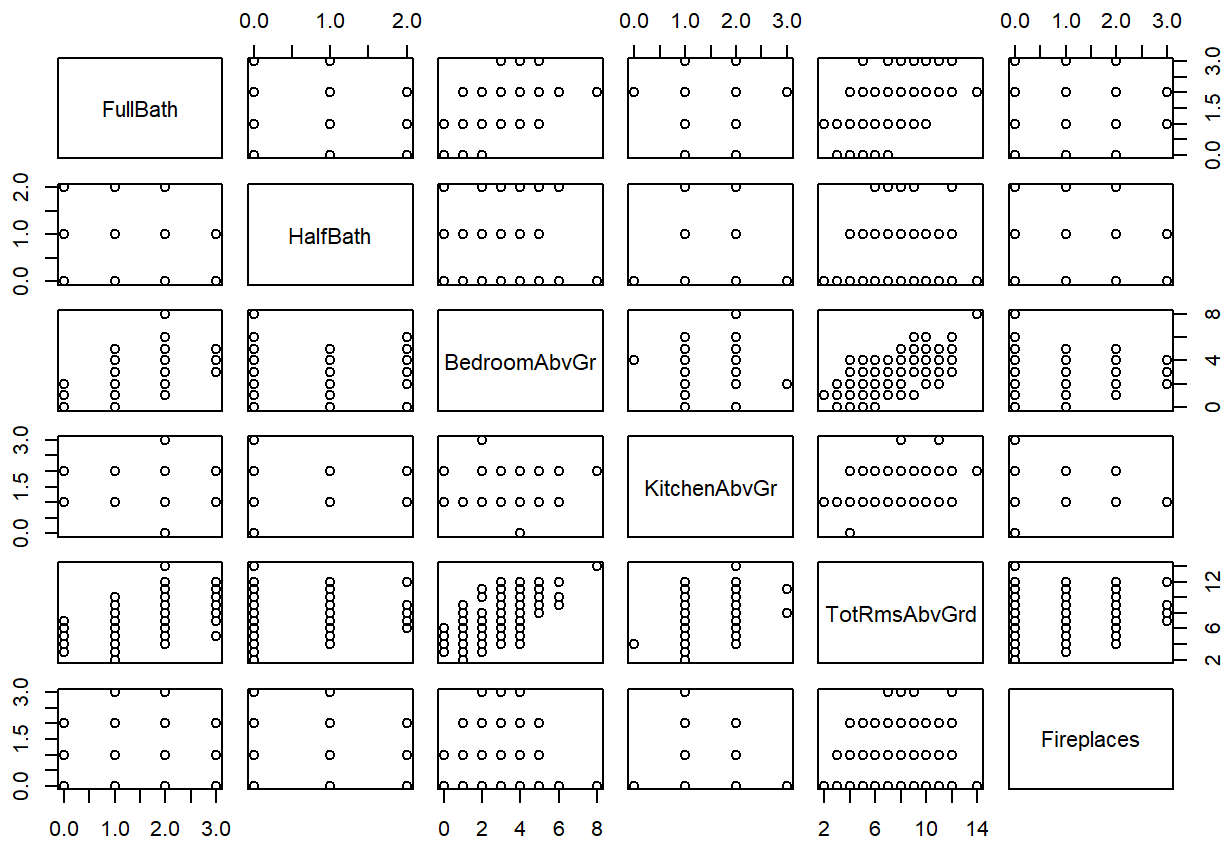
```
plot(datos_train[,5:10])
```



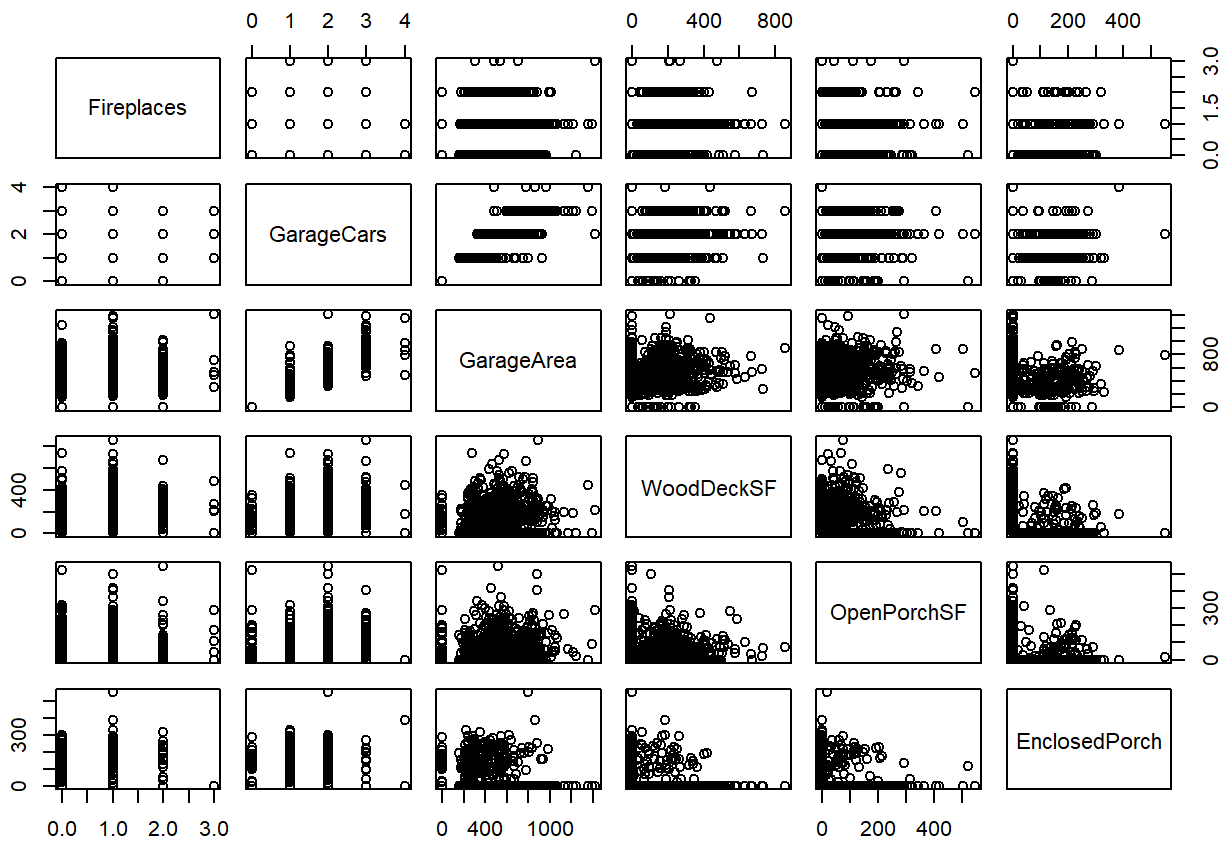
```
plot(datos_train[,10:15])
```



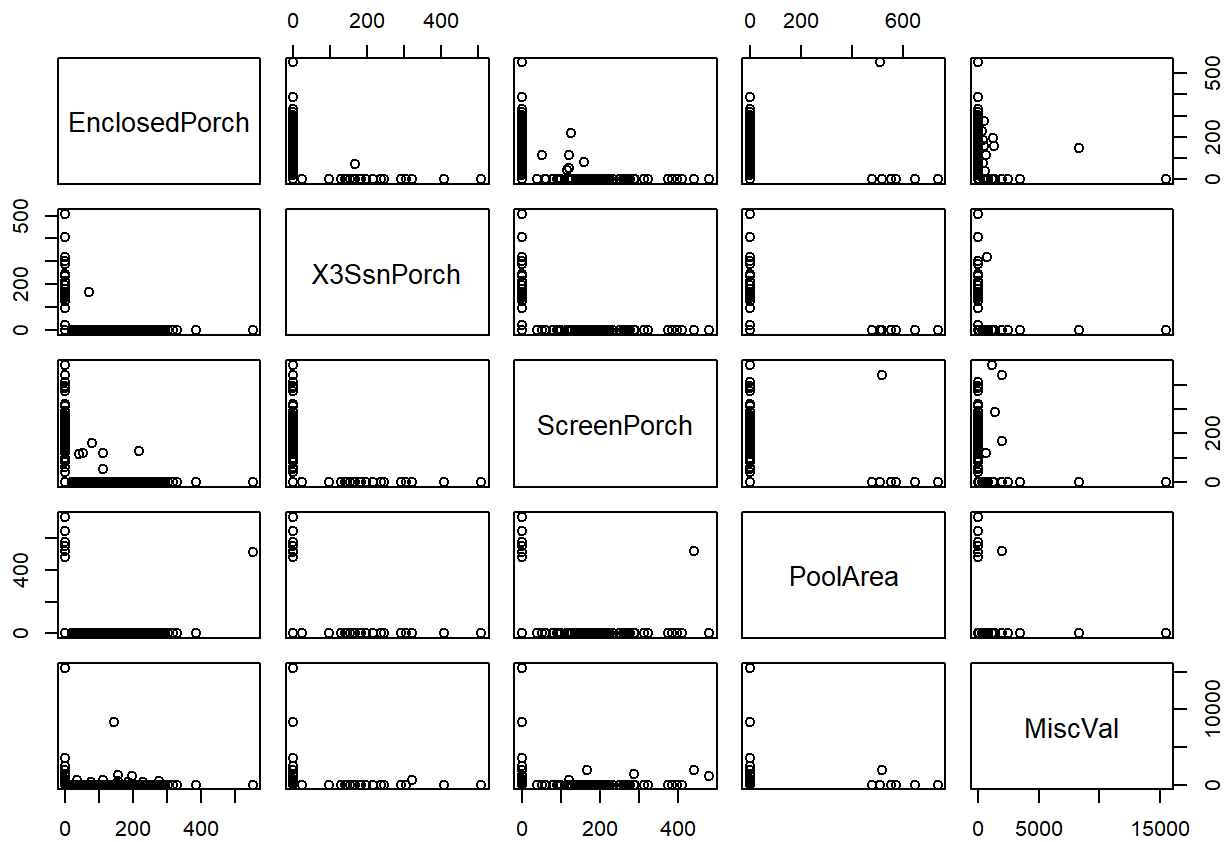
```
plot(datos_train[,15:20])
```



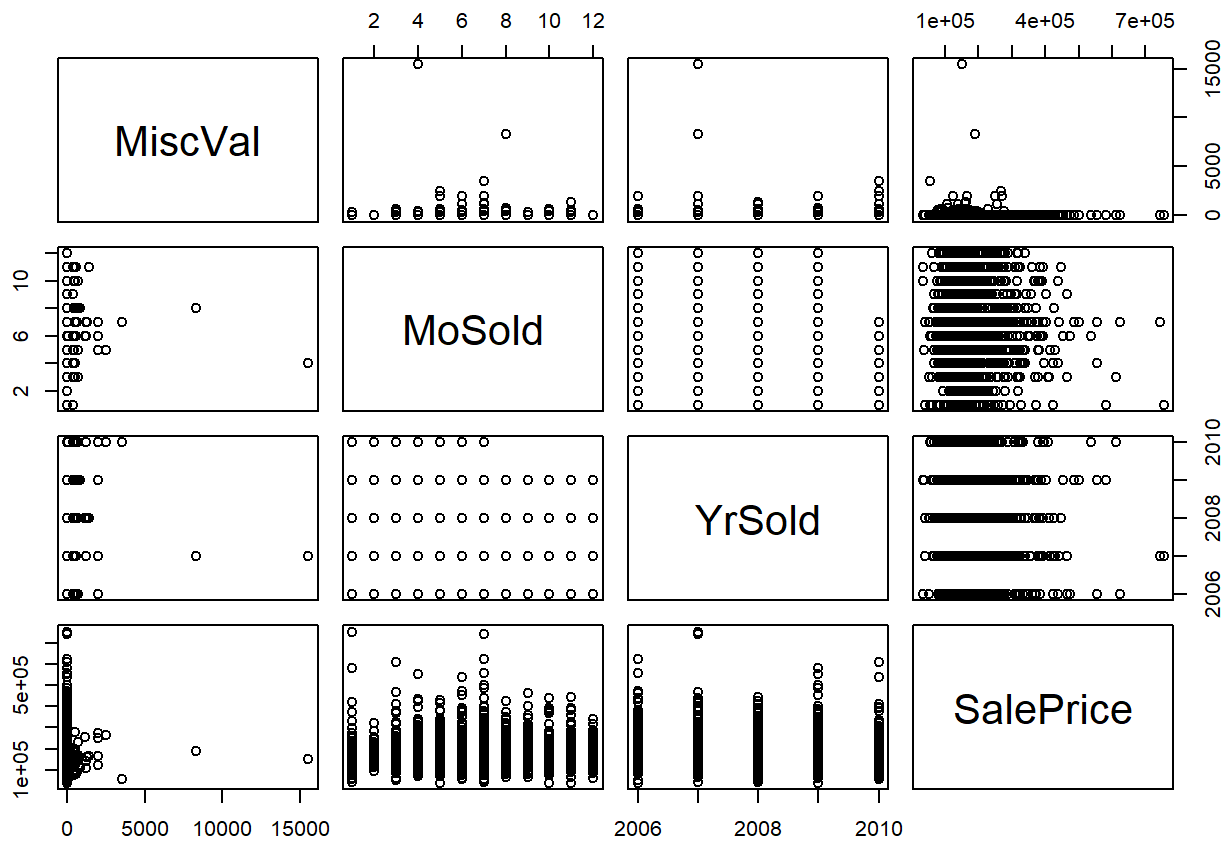
```
plot(datos_train[,20:25])
```



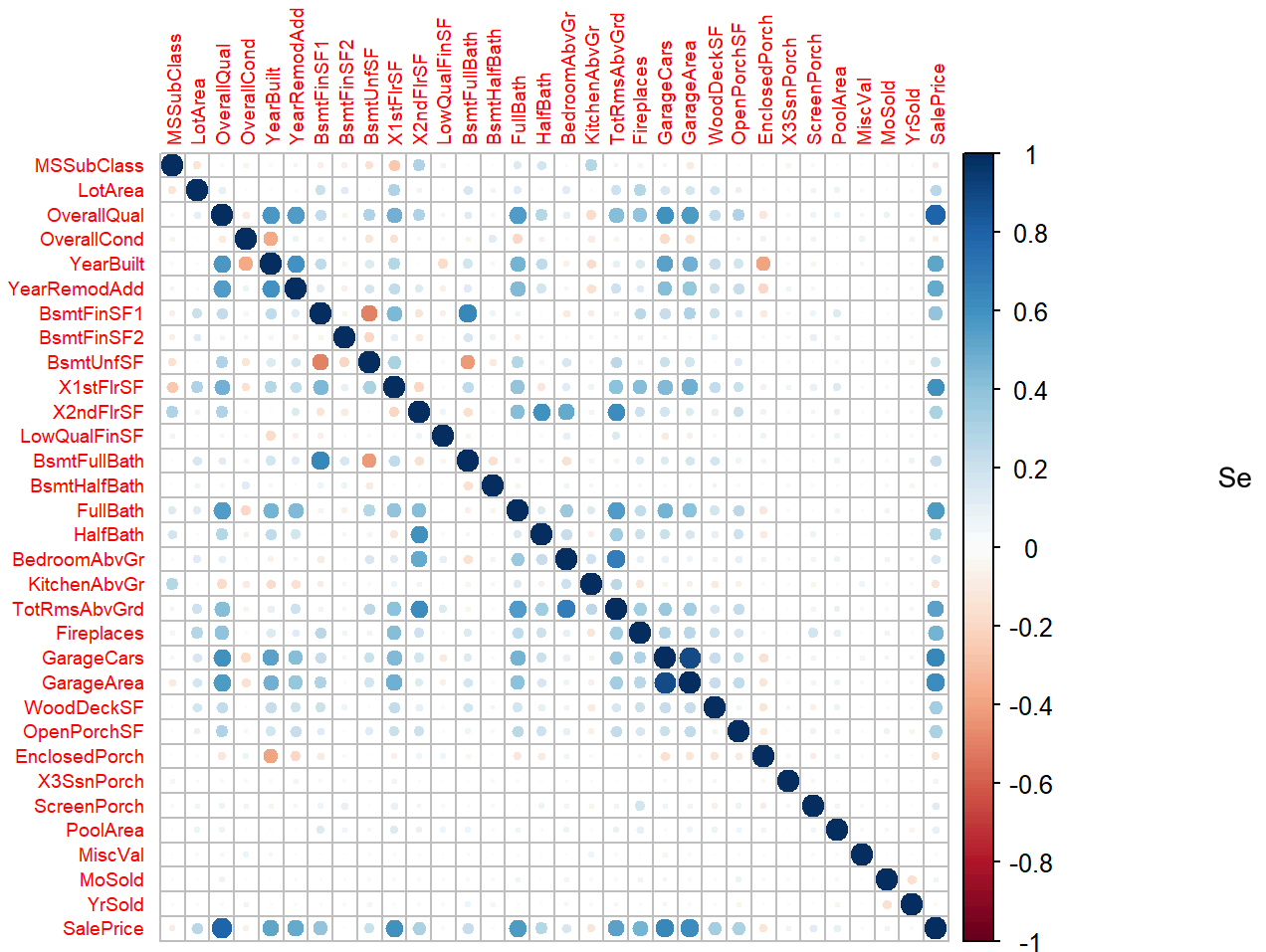
```
plot(datos_train[,25:29])
```



```
plot(datos_train[,29:32])
```



```
matriz_cor <- cor(datos_train)
corrplot(matriz_cor, tl.cex = 0.6)
```

observa que GarageCars, TotRmsAbvGrd y Garage Area se relacionan y ademas Sales price con Overall Quality exponen multicolinealidad ya que su correlación es mayor a 90%. Por otro lado se quita la variable MoSold para inferir si el R^2 ajustado es igual a 1.

Para esto se quita las variables GarageCars, Garage Area, TotRmsAbvGrd y Overall Quality. Sales price no se quita porque es la variable principal que comparamos.

```
datos <- datos[, c("MSSubClass", "LotArea", "OverallQual", "OverallCond",
  "YearBuilt", "YearRemodAdd", "BsmtFinSF1", "BsmtFinSF2",
  "BsmtUnfSF", "X1stFlrSF", "X2ndFlrSF", "LowQualFinSF",
  "BsmtFullBath", "BsmtHalfBath", "FullBath", "HalfBath",
  "BedroomAbvGr", "KitchenAbvGr", "Fireplaces",
  "WoodDeckSF", "OpenPorchSF", "EnclosedPorch",
  "X3SsnPorch", "ScreenPorch", "PoolArea", "MiscVal", "YrSold",
  "SalePrice")]

datos_train <- datos_train[, c("MSSubClass", "LotArea", "OverallQual", "OverallCond",
  "YearBuilt", "YearRemodAdd", "BsmtFinSF1", "BsmtFinSF2",
  "BsmtUnfSF", "X1stFlrSF", "X2ndFlrSF", "LowQualFinSF",
  "BsmtFullBath", "BsmtHalfBath", "FullBath", "HalfBath",
  "BedroomAbvGr", "KitchenAbvGr", "Fireplaces",
  "WoodDeckSF", "OpenPorchSF", "EnclosedPorch",
  "X3SsnPorch", "ScreenPorch", "PoolArea", "MiscVal", "YrSold",
  "SalePrice")]
```

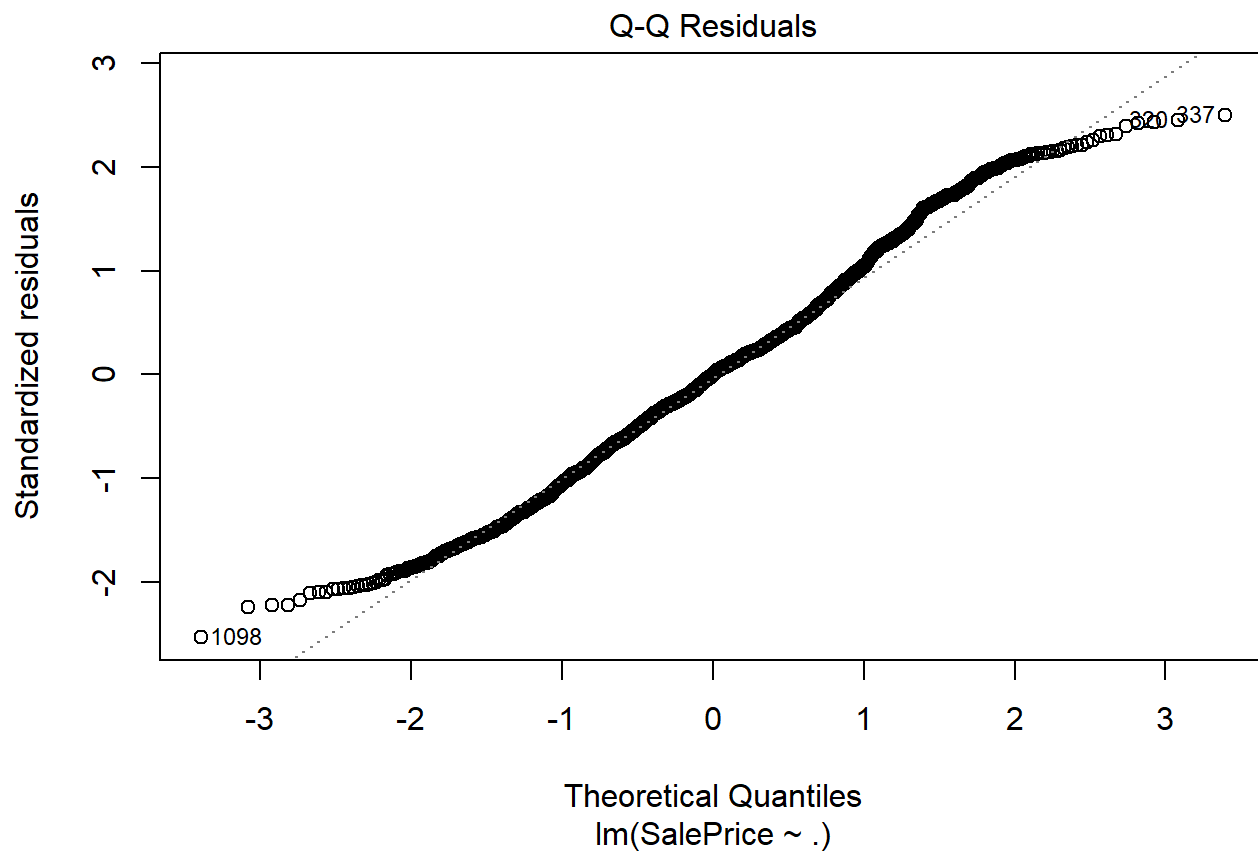
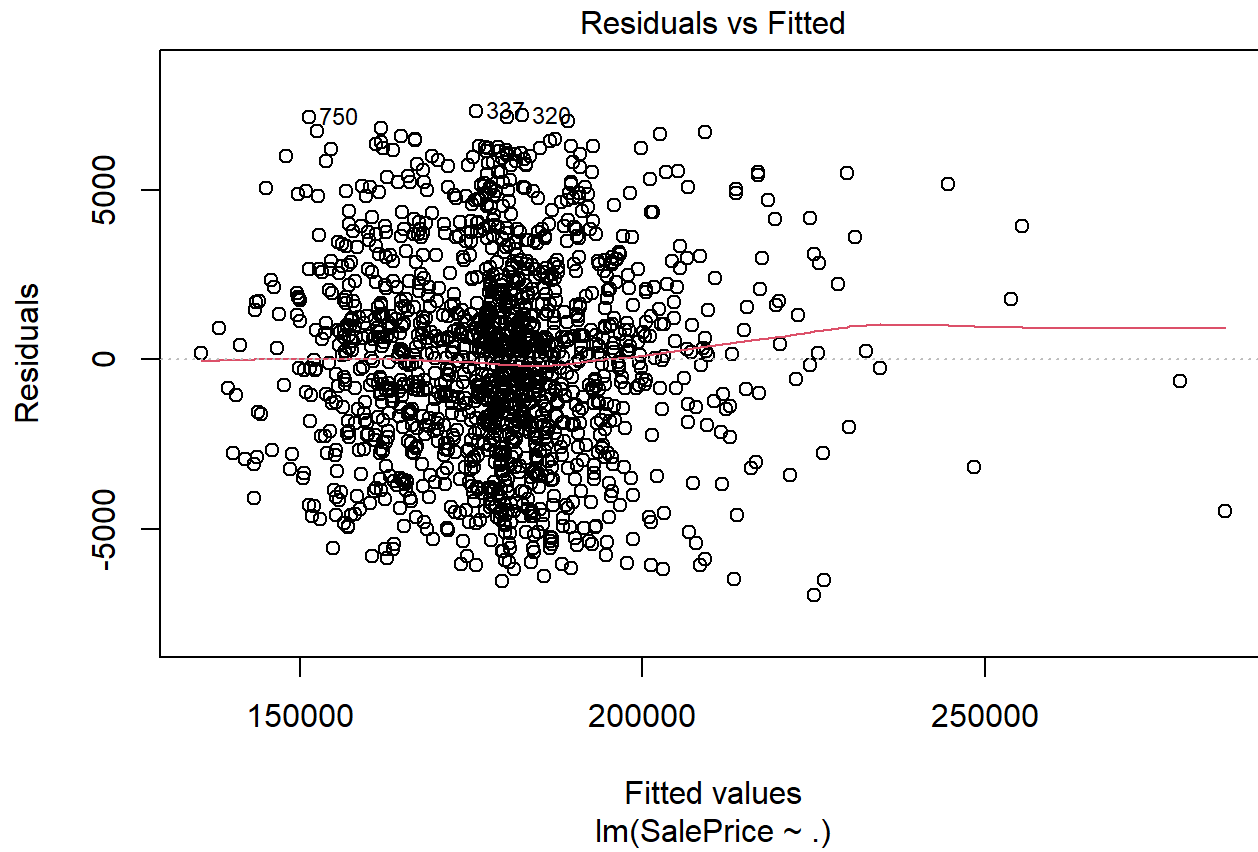
```
fitMLM_varSales_noVar<-lm(SalePrice~.,data = datos)
```

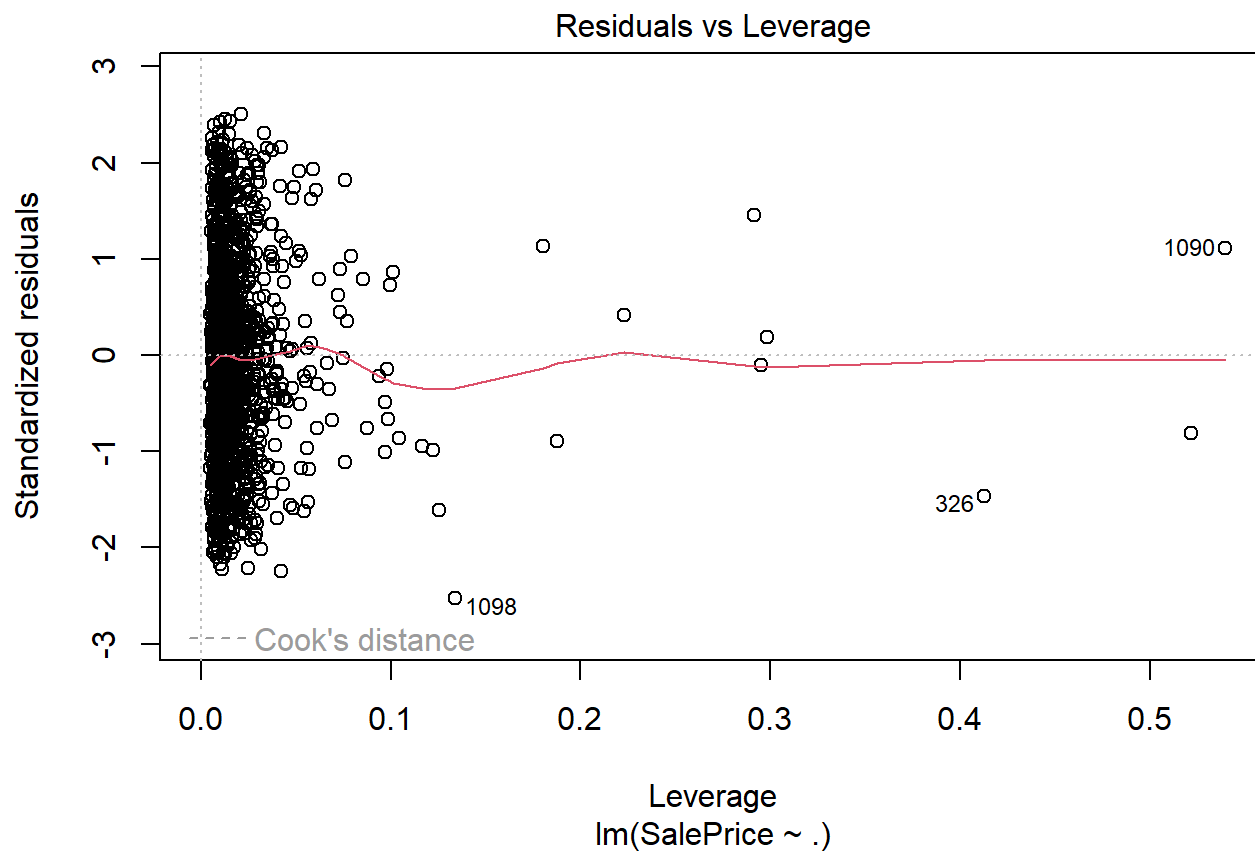
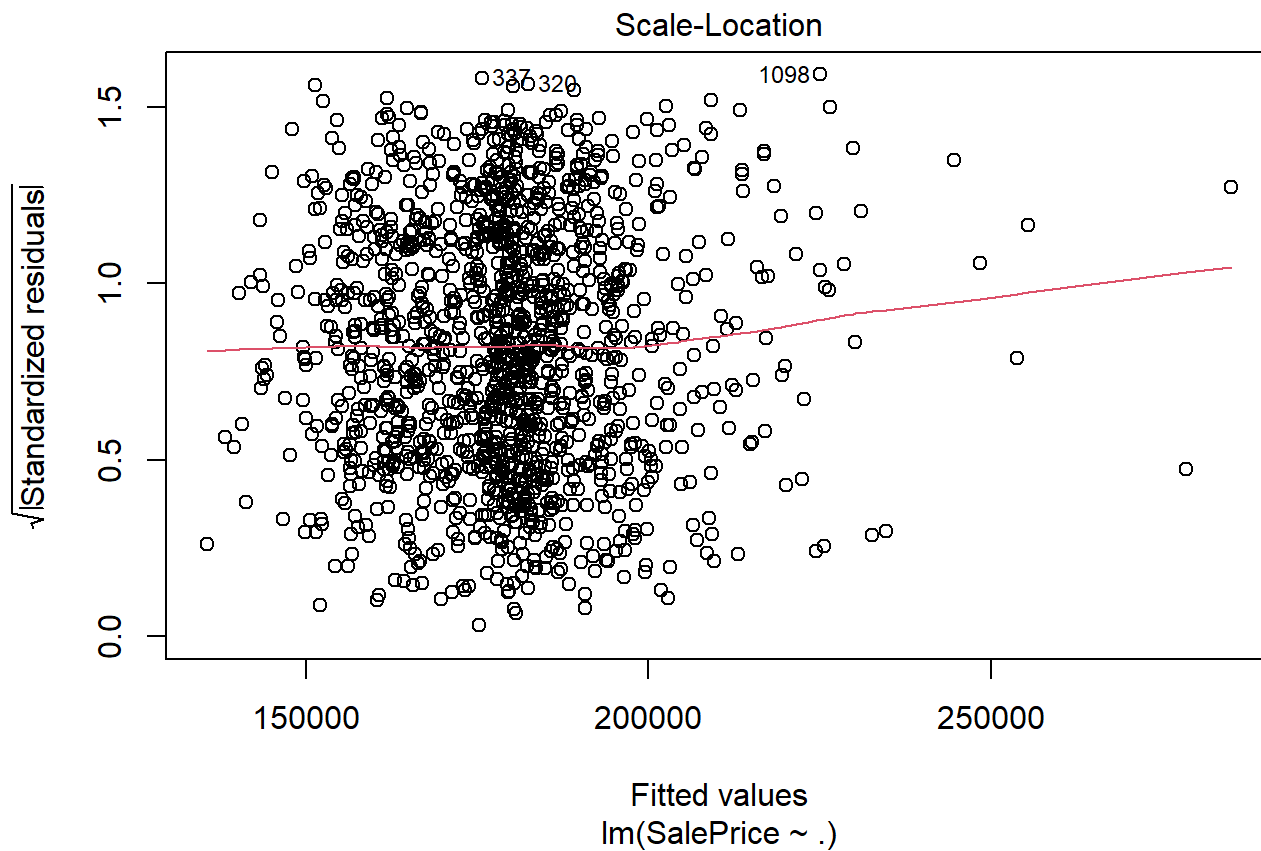
```
summary(fitMLM_varSales_noVar)
```

```
##
## Call:
## lm(formula = SalePrice ~ ., data = datos)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -6972.4 -1998.2   -18.3  1819.8  7327.8
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   2.720e+06  1.211e+05  22.464 < 2e-16 ***
## MSSubClass     9.422e-01  2.220e+00   0.424  0.67132
## LotArea       1.944e+00  2.010e-02  96.715 < 2e-16 ***
## OverallQual  -1.376e+02  9.512e+01  -1.446  0.14826
## OverallCond  -1.013e+01  8.545e+01  -0.119  0.90562
## YearBuilt     8.142e-01  4.812e+00   0.169  0.86567
## YearRemodAdd   5.560e+00  5.660e+00   0.982  0.32612
## BsmtFinSF1    -5.494e-01  3.914e-01  -1.404  0.16063
## BsmtFinSF2    -6.704e-01  5.699e-01  -1.176  0.23965
## BsmtUnfSF     -4.187e-01  3.463e-01  -1.209  0.22680
## X1stFlrSF     6.473e-01  4.449e-01   1.455  0.14585
## X2ndFlrSF    -5.365e-01  3.892e-01  -1.378  0.16831
## LowQualFinSF   3.153e+00  1.798e+00   1.754  0.07972 .
## BsmtFullBath   2.536e+02  2.126e+02   1.193  0.23314
## BsmtHalfBath   3.708e+02  3.387e+02   1.095  0.27383
## FullBath       1.529e+02  2.271e+02   0.673  0.50095
## HalfBath       2.228e+02  2.285e+02   0.975  0.32950
## BedroomAbvGr   1.360e+04  1.303e+02  104.308 < 2e-16 ***
## KitchenAbvGr   2.331e+02  4.482e+02   0.520  0.60317
## Fireplaces     9.489e+01  1.495e+02   0.635  0.52569
## WoodDeckSF     3.584e-01  6.856e-01   0.523  0.60129
## OpenPorchSF   -1.516e-01  1.266e+00  -0.120  0.90472
## EnclosedPorch   5.288e-01  1.292e+00   0.409  0.68227
## X3SsnPorch     4.678e+00  3.890e+00   1.202  0.22939
## ScreenPorch    1.821e+00  1.426e+00   1.277  0.20169
## PoolArea      -7.306e+00  2.711e+00  -2.695  0.00712 **
## MiscVal        1.000e-01  1.312e-01   0.763  0.44585
## YrSold        -1.300e+03  6.016e+01 -21.617 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2960 on 1429 degrees of freedom
## (2 observations deleted due to missingness)
## Multiple R-squared:  0.9676, Adjusted R-squared:  0.967
## F-statistic: 1583 on 27 and 1429 DF, p-value: < 2.2e-16
```

Se observa que R^2 ajustado y el multiple ya no es perfecto pero es razonable ya que es mayor a 90%

```
plot(fitMLM_varSales_noVar)
```

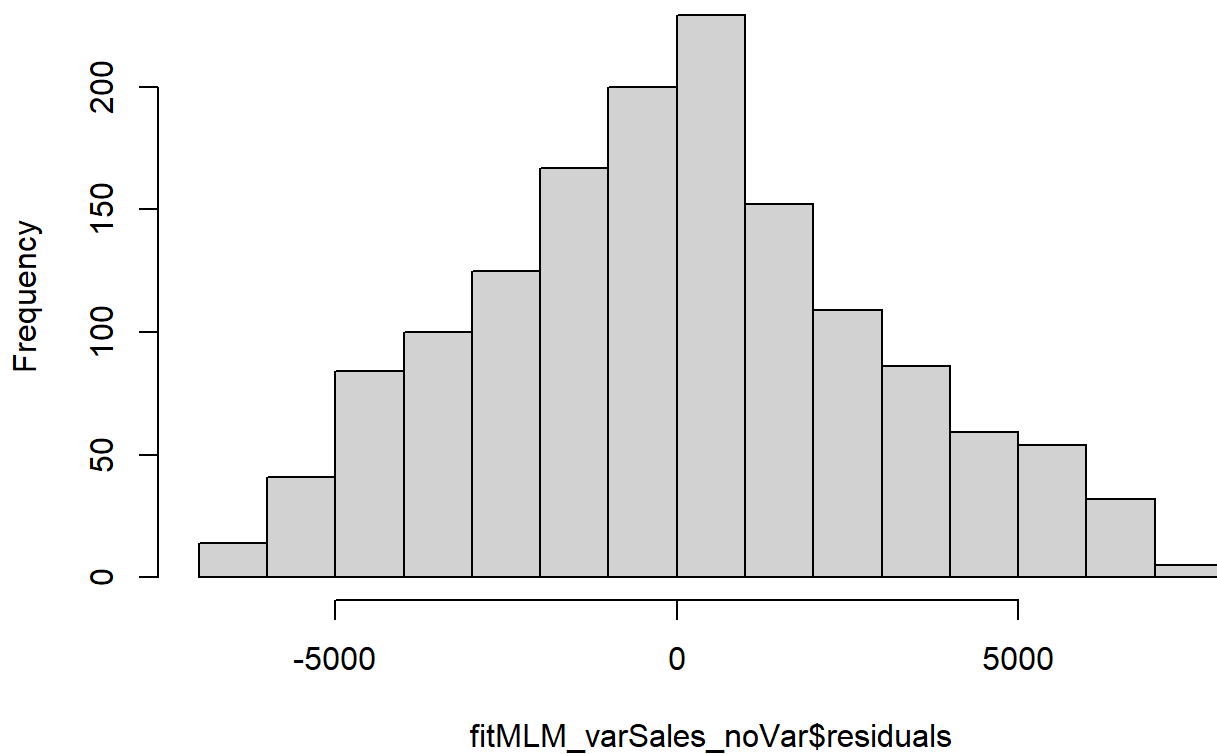




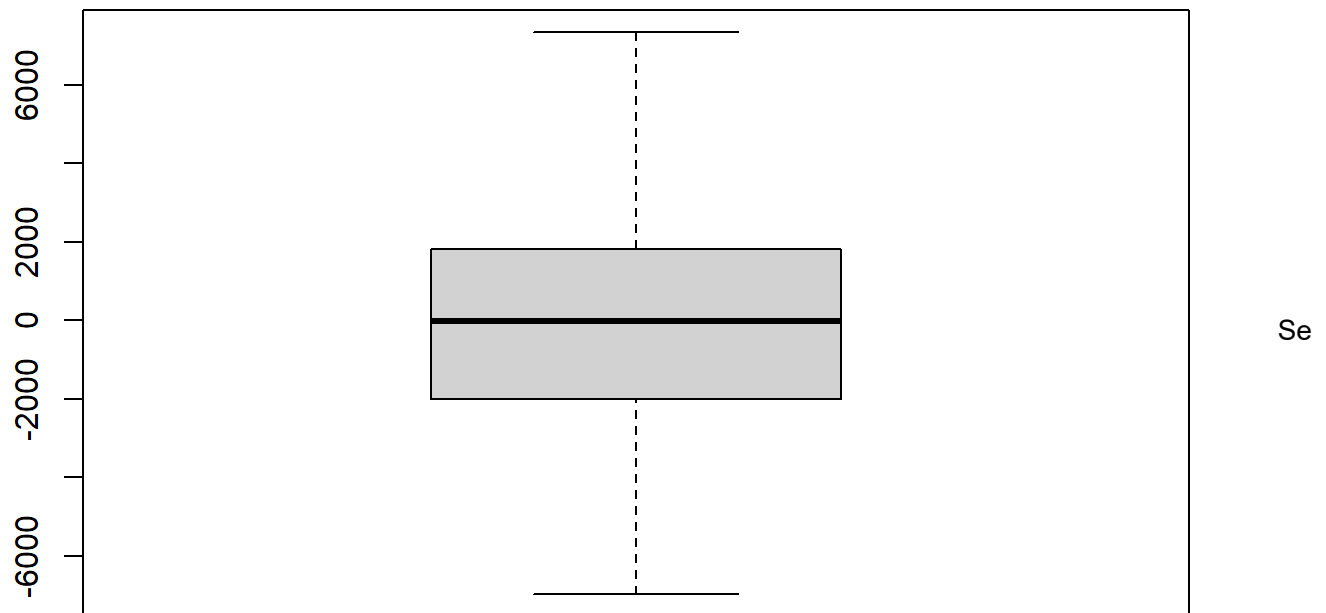
En los gráficos se pueden ver que los residuos están aleatoriamente distribuidos alrededor de 0 y que no siguen ningún patrón. Si bien el gráfico q-q parece que no siguen una distribución normal como se puede observa en las cola inicial y cola del final por esta razón, necesitamos realizar la prueba de Kolmogorov-Smirnov.

```
hist(fitMLM_varSales_noVar$residuals)
```

Histogram of fitMLM_varSales_noVar\$residuals



```
boxplot(fitMLM_varSales_noVar$residuals)
```



observa que en la caja y bigote no tiene puntos atípicos y la distribución se asimila pero no es normal.

```
lillie.test(fitMLM_varSales_noVar$residuals)
```

```
##
##  Lilliefors (Kolmogorov-Smirnov) normality test
##
## data:  fitMLM_varSales_noVar$residuals
## D = 0.03165, p-value = 0.001651
```

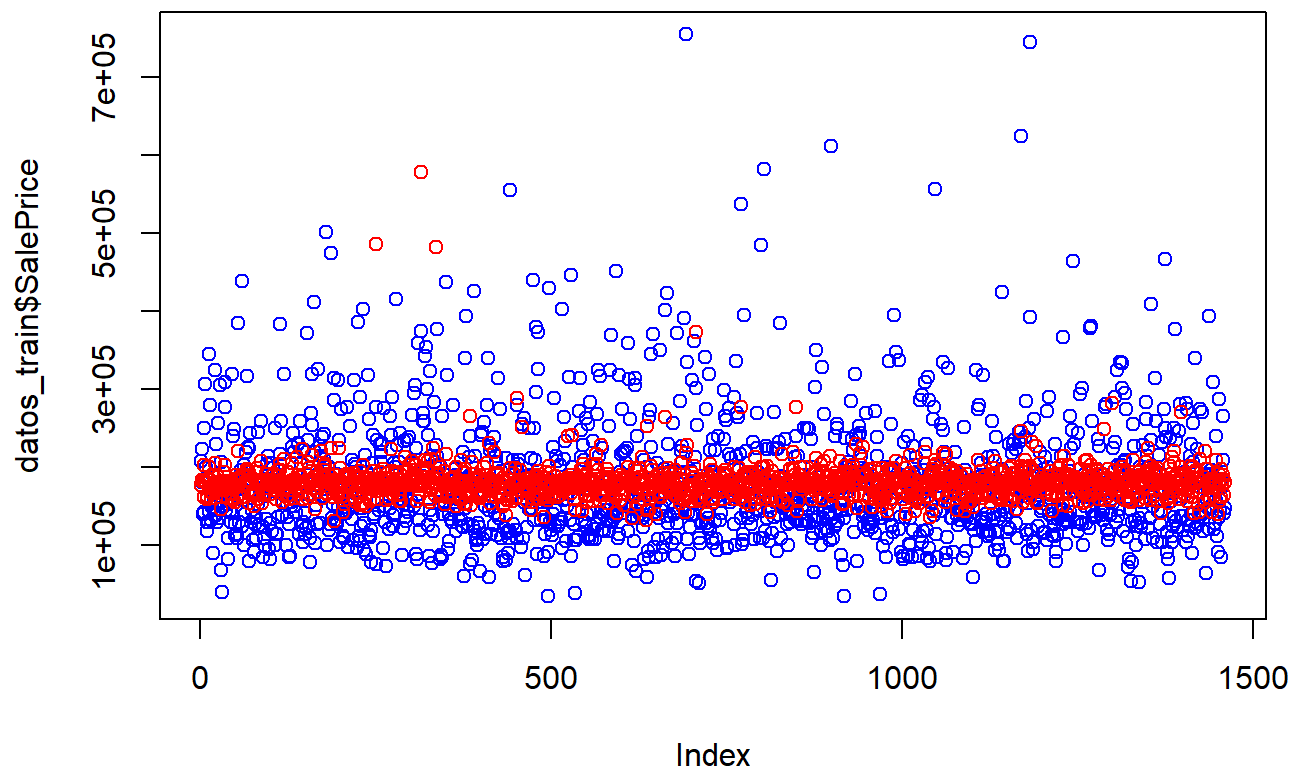
Se observa que el p-value es menor a 0.5 lo cual aceptamos la hipótesis nula donde la distribución no es normal.

```
predMLM<-predict(fitMLM_varSales_noVar,newdata = datos_train)
RMSE(predMLM,datos_train$SalePrice)
```

```
## [1] 75818.32
```

El RMSE es de 75,818.32 donde indica que, las predicciones del modelo se desvían del precio real de venta en esa cantidad. Si los precios de las casas son elevados, el error puede ser aceptable pero cuando los precios de las casas son bajos el error si puede afectar gravemente.

```
plot(datos_train$SalePrice,col="blue")
points(predMLM, col="red")
```



```
summary(datos_train$SalePrice-predMLM)
```

##	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
##	-253866.4	-48904.8	-17452.9	276.8	30982.4	542769.0

Se observa que los puntos rojos siguen una tendencia similar a los azules, lo que sugiere que el modelo captura en gran medida el comportamiento de los datos. Sin embargo, se observan algunas desviaciones, especialmente en valores altos, donde el modelo puede no estar prediciendo con precisión.